Vani Gupta

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EXPERIENCE

Oracle **Redwood City, CA**

Software Engineer I *Feb 2025 - Present*

- Sole engineer on a versioned publishing system: wrote the design doc, then shipped the schema, REST APIs and lifecycle state machine with transition validations
- Remodeled compliance findings from free-text blobs into a normalized relational schema (stable entity identity, document references, per-criterion anchors), with CRUD APIs and export changes shipped behind format-compatibility flags for a zero-downtime rollout. Made finding-level data queryable for the first time and replaced whole-document rewrites with in-place edits.
- Cut a quarterly dependency-resolution job from ~3 minutes to ~10 seconds across 1,000+ documents with transitive dependency graphs by batching writes, adding targeted indexes, and caching resolved state
- Built the platform's audit-log and rollback system: every bulk operation records actor, timing, and per-item outcome with a reversible changeset. Bounded storage via delta-only persistence and lazy changelog loading, replacing ~1 MB JSON payloads per action and cutting database memory load 60%.
- Built the service's authorization layer with SRE; primary on-call : caught a production CrashLoopBackOff on a core identity service within minutes and resolved it within the hour; Ran zero-downtime migrations on live production tables including a ~10,000-record migration off a deprecated legacy tool

Software Engineer Intern *May 2024 - Aug 2024*

- Reduced pre-deployment validation from over 10 hours to roughly 10 minutes across ~20 microservices by building automated vulnerability scanning, end-to-end test orchestration, and scheduled CI/CD pipelines.
- Deployed and operated containerized services on Oracle Kubernetes Engine, supporting production-grade infrastructure with Terraform, Ansible, and Oracle Cloud.

Software Engineer Intern *May 2023 - Aug 2023*

- Built an internal Node.js application automating sales-demo provisioning previously assembled by hand in Excel.
- Designed the backend microservices (Node.js, Knex/Objection.js) on Oracle Kubernetes Engine, cutting read-path response times by consolidating repeated database round trips into shared data-access modules and stored procedures.
- Productized the application into Oracle Spectra; Owned the non-prod to prod release path; Designed the deployment architecture, and built the dataset layer backing the OpenTofu provider.

PROJECTS

Kafka compatible event-streaming broker in Rust

- Follows the Apache Kafka wire protocol, so unmodified kcat clients produce and consume against it: 12 API keys, record batch v2 with CRC-32C, consumer groups with the full rebalance protocol, and leader-follower replication with ISR tracking and high-watermark reads.
- Segmented append-only log with a sparse offset index (8 B per 4 KiB of log); binary search plus a bounded forward scan makes lookup cost independent of log size. 898k msg/s produce, 423k msg/s consume on an M2.
- Built a deterministic crash simulator behind a storage trait that models power loss and partial persistence from a 64-bit seed, so any failure reproduces exactly; 2,000 seeded runs × 3 crash cycles assert the log is always a contiguous prefix of acknowledged writes, with mutation tests proving the harness can actually fail.

Multi-Agent Debate Framework for SLM Reasoning ***Published in ACM ICAIF 2026***

- Fine-tuned Gemma models via SFT/LoRA into Quant, Fundamental, Sentiment, and Economist agents on a synthetic financial reasoning corpus generated with Qwen-32B.
- Architected a LangGraph multi-agent system with adversarial review, moderator-guided debate, and structured output validation, feeding portfolio views into Black-Litterman optimization.

SKILLS

Languages/DB/Frameworks: Python, C++, Java, Rust, React.js, Node.js, OracleDB, PostgreSQL, MongoDB, SQL

Infrastructure & Dev Tools: Docker, Kubernetes, Terraform, Ansible, AWS, Oracle Cloud, Grafana, Postman, Redis, gRPC

ML/AI: PyTorch, CUDA, OpenCV, LLM Fine-tuning, RAG, LangGraph

EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY **Atlanta, GA**
Online M.S. in Computer Science *Jan 2026 - Dec 2027*

UNIVERSITY OF MASSACHUSETTS AMHERST **Amherst, MA**
B.S. in Computer Science *Sep 2021 - Dec 2024*